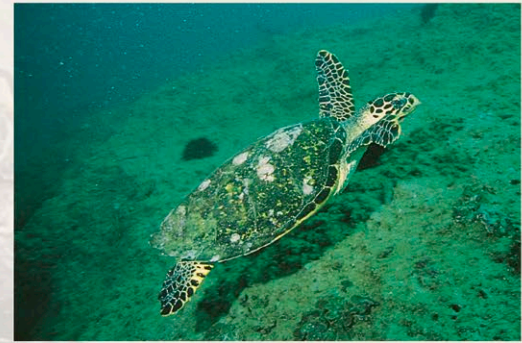


Migration

Some sea turtles migrate long distances from their feeding grounds to the beaches where they lay their eggs. For example, some green sea turtles feed off the coast of Brazil and nest on Ascension Island, in the eastern Atlantic, which involves a return journey of at least 2,800 miles (4,500 km). All the breeding adults arrive at the nesting beaches within a few weeks of each other. Exactly how they navigate is still poorly understood, but they probably use a combination of the earth's magnetic field, the directions of ocean currents, water chemistry, and memory. The greatest distances are traveled by the leatherback turtle, which ranges from tropical seas almost to Arctic waters in pursuit of jellyfish, its preferred food.



SWIMMING

When swimming, sea turtles (such as this hawksbill turtle) use their front limbs for propulsion and their back limbs for steering. The speed at which they swim varies between species, from 1½ to about 18 mph (3 to 30 kph).

Reproduction

All tortoises and turtles lay their eggs on land. Tropical species may lay several clutches throughout the year, whereas temperate species lay only once or twice during a breeding season; females can store sperm so that they can continue to lay fertile eggs long after mating. The size of the clutch depends on the size of the species: small turtles lay 1–4 eggs, while the largest species can lay over 100. The green sea turtle is the most prolific species, laying up to 6 or 7 clutches of more than 100 eggs spread over 2 weeks. Turtles do not care for their young, although at least one species (the Burmese brown tortoise) stays at its nest for a few days after laying to protect the eggs from predators.



EGGS

The eggs of large turtles and tortoises are almost spherical, but those of small species tend to be elongated. The shells may be either hard and brittle or flexible. This leopard tortoise hatchling is breaking out of its shell using an egg tooth that will be shed shortly afterward.

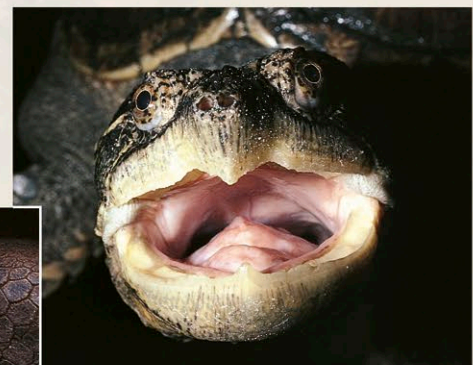


NESTING

Some female tortoises and turtles lay their eggs under rotting vegetation or in the burrows of other animals. However, most species (such as this green sea turtle) use their back limbs to dig a special flask-shaped chamber. Once the eggs have been laid, the sand or soil is carefully replaced and smoothed over to conceal them from predators.

FRESHWATER CARNIVORES

Snapping turtles live in shallow lakes, rivers, and swamps. This common snapping turtle (right) will prey on almost anything that comes within range of its sharp jaws and cavernous mouth, and is small enough to be swallowed.



VEGETARIANS

Tortoises such as this Galapagos tortoise (left) graze on grass and other low plants, or browse on bushes and shrubs. They often eat almost continuously when active and are quick to take advantage of more nutritious fallen fruit and even animal carcasses if they are available.



Feeding

Turtles and tortoises are too slow-moving to pursue active prey, although a few aquatic species—mostly those that live in murky, clouded water—hunt by ambush, remaining motionless in the hope that a fish or crustacean will inadvertently swim past. Tortoises are primarily herbivorous, grazing or browsing on leaves and fruit. Many species also eat animals, such as caterpillars, that appear incidentally in their food, and most will eat carrion given the chance. Freshwater turtles often start life as insectivores, finding enough small aquatic larvae and other small prey to survive but, as they grow, they switch to a diet consisting mainly of aquatic vegetation. Some marine turtles eat only seaweed as adults, while others feed on invertebrates, including jellyfish, sea urchins, and mollusks.